

Multi-Round Belief Propagation Decoding With Impulsive Perturbation for Short LDPC Codes

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Abstract—The belief propagation (BP) decoding of short low-density parity check codes suffers from short cycles and trapping sets. A number of perturbation schemes have been proposed to overcome the limitation. This letter presents a generalized framework for multi-round BP decoding with input perturbation and proposes an improved perturbation scheme. In particular, the impulsive perturbation which is applied to only a few symbols is proposed, along with a new space selection criterion. The proposed scheme performs better than existing schemes for short codes at lower complexity and supports a parallel implementation.

Index Terms—Low-density parity check codes, belief propagation decoding, perturbation-based decoding, ultra-reliable and low latency communication.

I. INTRODUCTION

LOW-DENSITY parity check (LDPC) codes are a class of capacity approaching codes, which have been adopted as a channel code for many wireless standards such as wireless local area network, digital video broadcasting, and recent 5G New Radio, etc [1]. Owing to their excellent error correcting performance and the availability of super-fast decoders, LDPC codes are also one of strong candidates for the channel code of ultra reliable low latency communications [2]. Even though LDPC codes along with their standard belief propagation (BP) decoding comply with the requirements of many applications, the unique conditions such as short block lengths, no error floor, and very low latency, provide new and different challenges to the study of LDPC codes [2]. To that end, we specially aim to improve the decoding performance of short LDPC codes in this letter.

Since BP decoding can be analyzed well as the code length goes to infinity [3], the design of LDPC codes has been constituted in two phases; ensemble design and code construction. This design manner has been very effective for sufficiently long codes. However, the BP decoding may suffer

from message dependency for short codes and get degraded from the optimal performance [4], which means there is a margin for improvement in decoding performance. Therefore, some modified or advanced BP-based decoding methods have been proposed to overcome the shortcomings [5]–[9], that provide a substantial coding gain. We refer to those algorithms as the multi-round BP decoding with perturbation (MBPDP). MBPDP schemes apply a type of perturbation to the received signal and conduct BP decoding iteratively.

In this letter, we first present a generalized MBPDP framework that can include a majority of MBPDP schemes. Then, we propose a new perturbation scheme called the ‘impulsive perturbation,’ which applies perturbation on only one or a few symbols toward the opposite direction to their values. Since the number of symbols perturbed is small, the search for a successful perturbation can be conducted in a wider space. A new perturbation space selection criterion that utilizes the number of mismatches of the extrinsic and posterior log likelihood ratio (LLR) is also proposed. Numerical results indicate that the proposed impulsive perturbation and space selection criterion outperforms existing MBPDP schemes for short LDPC codes at a slightly lower complexity.

II. BP DECODING WITH PERTURBATION

We assume an (n, k) binary LDPC code $\mathcal{C} = \{c | cH^T = 0\}$ defined by a binary parity check matrix H . We consider codeword $c = (c_1, \dots, c_n) \in \{0, 1\}^n$ and its binary phase shift keying (BPSK) modulated signal $x = (x_1, \dots, x_n)$ with $x_i = (-1)^{c_i}$. A cyclic redundancy check (CRC) code is concatenated with the LDPC code for error detection. Let $r = (r_1, \dots, r_n) = x + n \in \mathbb{R}^n$ be the received signal, where n is the additive white Gaussian noise (AWGN) with entries independent identically distributed with zero mean and variance σ^2 . Let $l = (l_1, \dots, l_n)$ be the intrinsic LLR of c determined by r . The decoder outputs a codeword estimate $\hat{c} = (\hat{c}_1, \dots, \hat{c}_n)$.

A. Multi-Round BP Decoding With Perturbation (MBPDP)

The MBPDP schemes commonly perform a conventional BP decoding with input perturbation until it finds a valid codeword, where BP decoding could be the generic iterative sum-product decoding, the min-sum decoding, or their modifications. When a full BP decoding fails, i.e., the syndrome check is unsatisfied, a new BP decoding round starts after perturbation of the received signal. The perturbation is the process of adding synthetic noise to the received signal. This process is repeated so as to reach a CRC-passing codeword.

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Algorithm 1 Generalized MBPDP

Input: $\mathbf{r}$, T , τ , and perturbation mode

- 1: Calculate the intrinsic LLR, the decoder input $\mathbf{l} = \frac{2}{\sigma^2} \mathbf{r}$
- 2: **for** $t = 1 : T + 1$ **do**
- 3: Perform a BP decoding with I_{max} and CRC check
- 4: **if** LDPC syndrome and CRC check are satisfied **then**
- 5: **return** Codeword $\hat{\mathbf{c}}$
- 6: **else if** $t = T + 1$ **then**
- 7: **return** ‘decoding failure’
- 8: **else**
- 9: **if** Perturbation space is dynamic or $t = 1$ **then**
- 10: Determine the order $\pi(\cdot)$ and $\mathbf{R}$
- 11: **end if**
- 12: **if** Perturbation mode is random **then**
- 13: Generate $\mathbf{z} = (z_1, \dots, z_\tau)$ and $\mathbf{r}' \leftarrow \text{Pert}(\mathbf{r}, \mathbf{z})$
- 14: **else if** Perturbation mode is deterministic **then**
- 15: **if** $t = 1$ **then**
- 16: Generate $\mathcal{P} = \{\mathbf{p}^1, \dots, \mathbf{p}^T\}$
- 17: **end if**
- 18: $\mathbf{r}' \leftarrow \text{Pert}(\mathbf{r}, \mathbf{p}^t)$
- 19: **end if**
- 20: Calculate new decoder input $\mathbf{l}' = \frac{2}{\sigma^2} \mathbf{r}'$
- 21: **end if**
- 22: **end for**

Whenever the syndrome check in the BP decoding is satisfied, a CRC check is conducted for the decoding result. We consider the perturbation as a modification of several received symbols in $\mathbf{r}$, which is undertaken only if LDPC syndrome or CRC check are unsatisfied. Let T be the maximum number of perturbation rounds. In the first step of perturbation, a subspace called the perturbation space is selected from the received signal space by collecting a set of coordinates. Let $\mathbf{R}$ be an ordered index set of the coordinates of the perturbation space of dimension τ . $\mathbf{R}$ is selected according to a criterion that reflects the reliability of the received symbols, since it is desired to perturb symbols with low reliabilities. Different criteria such as intrinsic LLR magnitude [7], the number of unsatisfied check nodes (nUC) [5], and the number of sign changes in the variable node messages (nSCV) [9] or a combination of them can be used to assess the reliability. Let λ_i be a quantified reliability of the i -th received symbol and $\pi(\cdot)$ be the index permutation such that $\lambda_{\pi(1)} \leq \lambda_{\pi(2)} \leq \dots \leq \lambda_{\pi(n)}$. Then we have $\mathbf{R} = \{\pi(1), \dots, \pi(\tau)\}$ to which perturbation is applied. $\mathbf{R}$ can be static or dynamic during the entire MBPDP. For dynamic perturbation space, $\mathbf{R}$ is updated based on the BP decoding result and perturbed signal of each perturbation round, similar to [8]. In general, the dynamic perturbation space requires additional computational complexity and makes parallel implementation impossible.

Perturbation schemes can be classified into random and deterministic perturbations. First, random perturbation can be represented as

$$r'_{\pi(i)} = r_{\pi(i)} + z_i, 1 \leq i \leq \tau,$$

where r'_i is the i -th symbol after perturbation and z_i is the perturbation noise. Gaussian perturbation [5] uses $z_i \sim \mathcal{N}(0, \sigma_p^2)$ and binary random perturbation [6] is realized by $z_i = A(1 - 2x_i)$, $x_i \sim \text{Bern}(0.5)$ where $\text{Bern}(a)$ denotes a Bernoulli trial with probability a and A is a constant. At

Algorithm 2 Perturbation Pattern Generation

Input: T

Output: $\mathcal{P} = \{\mathbf{p}^1, \dots, \mathbf{p}^T\}$

- 1: $t \leftarrow 1$, $w \leftarrow 1$, $\mathbf{p}_{\text{new}} \leftarrow (0, \dots, 0)$
- 2: **while** $t \leq T$ **do**
- 3: Find a $\mathbf{p}_{\text{new}}$ of Hamming weight w next to $\mathbf{p}^{t-1}$ in lexicographical order
- 4: **if** $\mathbf{p}_{\text{new}}$ is found **then**
- 5: $\mathbf{p}^t \leftarrow \mathbf{p}_{\text{new}}$, $t \leftarrow t + 1$
- 6: **else**
- 7: $w \leftarrow w + 1$
- 8: **end if**
- 9: **end while**

each perturbation round, $\mathbf{r}' = (r'_1, \dots, r'_n)$ with the symbols outside $\mathbf{R}$ unchanged is used in the following BP decoding. In random perturbation, the pattern of perturbation is determined randomly. Decoding succeeds when the random sampling picks a proper perturbation pattern. In contrast, deterministic perturbation is designed with a set of perturbation patterns, $\mathcal{P} = \{\mathbf{p}^1, \dots, \mathbf{p}^T\}$, where $\mathcal{P}$ is an ordered set of binary τ -tuples. The afterburner [7] belongs to this class and is represented using $\mathbf{p} = (p_1, \dots, p_\tau)$ as

$$r'_{\pi(i)} = A(-1)^{p_i}, 1 \leq i \leq \tau,$$

where A is a constant, e.g., the maximum representable value. The received symbols are replaced by the perturbation pattern. In [7], afterburner uses $\mathcal{P}$ that includes all binary vectors of length τ . If T is preset, the size of perturbation space $\tau = \log_2 T$ is determined to be relatively small. To include an effective perturbation that leads to a successful decoding with high probability, $\mathbf{R}$ should be properly determined. The final step of perturbation is the preparation of the actual input to BP decoding; the modified LLR $\mathbf{l}' = \frac{2}{\sigma^2} \mathbf{r}'$.

B. Generalized MBPDP Scheme

In this subsection, we present a generalized framework that includes several existing MBPDP schemes [5]–[9] and our proposed scheme as a special case. The generalized MBPDP is described in Algorithm 1. First, a generic BP decoding with the maximum number of iterations I_{max} is performed. If the first round of BP decoding trial fails, then the perturbation decoding phase begins. The perturbation space is determined based on the space selection criteria. After perturbation patterns are planned, re-decoding is conducted with the received signal modified by a perturbation. The perturbation operation that generates the perturbed vectors $\mathbf{r}'$ is denoted by $\text{Pert}(\mathbf{r}, \cdot)$, where the second argument is a noise pattern $\mathbf{z}$ for random perturbation and a binary perturbation pattern $\mathbf{p}$ for deterministic perturbation. Maximally, T rounds of perturbation decoding are tried until a CRC-pass. In the absence of a CRC code, the most likely codeword is chosen among the list of candidates obtained after T rounds without early stopping. Performance behaviors are similar although the average computational complexity is high.

It is worth to mention the relationship between the generalized MBPDP and an important related work, ‘augmented BP

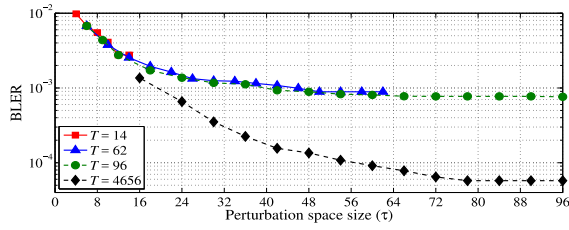


Fig. 1. BLER versus τ for fixed T : (96, 48) code, $E_b/N_0 = 3.50$ dB.

(AugBP)' [8]. AugBP can be understood as the iterative afterburner with growing perturbation space. Because AugBP can be considered as a combination of afterburner and dynamic perturbation space selection, we regard this scheme as an extension of the MBPDP framework. The algorithm has been improved with the nSCV criterion [9]. Thus, it is referred to as the improved AugBP (IAugBP) and taken as a performance benchmark.

III. NEW MBPDP SCHEME

A. Multi-Round BP Decoding With Impulsive Perturbation

Random perturbation can be regarded as a Monte-Carlo type search for finding a perturbation pattern which leads to a successful decoding. Because the random sampling takes place in a large space, a large number of trials are required to ensure finding a good one with high probability. It was also observed that the performance of the random perturbation degrades if a static perturbation space is used. On the other hand, a deterministic perturbation scheme, afterburner [7] manages a small perturbation space for limiting search space and attempts all possible perturbation patterns. In those approaches, the number of all perturbation patterns increases exponentially with the perturbation space size τ . So for the same volume of search space T , perturbation space is determined small. We suspect that the performance is limited because of that reason.

We consider that a perturbation noise can be tailored as one with few nonzero entries with a large perturbation value and the opposite sign of the received symbol, because the constructive perturbation noise to the received symbol gives a less impact on decoding than one with the opposite sign. The simplification converts the perturbation pattern search into a binary pattern search with which the effective search space can be increased for the same complexity. The proposed perturbation, which we call 'impulsive perturbation' applies to the prechosen perturbation space associated with $\mathbf{R}$ as

$$r'_{\pi(i)} = \begin{cases} -\text{sgn}(r_{\pi(i)})A_{\max}, & \text{if } p_i = 1, \\ r_{\pi(i)}, & \text{otherwise,} \end{cases}$$

for $1 \leq i \leq \tau$ for a binary perturbation pattern $\mathbf{p} \in \{0, 1\}^\tau$ and $\pi(i)$ is the index order of symbol reliability. In our approach, we intend to search for a successful perturbation pattern over an approximate Hamming sphere of a small radius in $\{0, 1\}^\tau$ associated with the perturbation space. In other words, $\mathcal{P}$ is composed of only $\mathbf{p}$'s with a small Hamming weight. Pattern $\mathbf{p}$'s are listed in the order of Hamming weight from smallest to largest. Among $\mathbf{p}$'s of the same weight, a lexicographical ordering is applied. The generation of $\mathcal{P}$ is realized as Algorithm 2. Let s be the maximum weight of $\mathbf{p} \in \mathcal{P}$, which

TABLE I
PERTURBATION PARAMETERS FOR SIMULATION

Parameters	Space update	Space selection	(96,48) LDPC ($T, s, \tau, A(\sigma_p)$)	(192,96) LDPC ($T, s, \tau, A(\sigma_p)$)
Proposed	Static	nSMV,	(14,1,14,max)	(30,1,30,max)
		LLR	(62,1,62,max)	(126,1,126,max)
		nSCV, LLR	(62,1,62,max)	(126,1,126,max)
IAugBP	Dynamic	nSCV,	(14,3,3,max)	(30,4,4,max)
		LLR	(62,5,5,max)	(126,6,6,max)
Afterburner	Static	LLR	(16,4,4,max)	(32,5,5,max)
			(64,6,6,max)	(128,7,7,max)
Gaussian	Dynamic	nUC, LLR	(14,12,12,0.6)	(30,36,36,0.5)
			(62,12,12,0.6)	(126,36,36,0.5)

is the radius of the smallest Hamming sphere that includes the entire $\mathcal{P}$. When the maximum number of perturbation rounds T is given, s is determined by the value of τ as

$$\sum_{i=1}^{s-1} \binom{\tau}{i} \leq T \leq \sum_{i=1}^s \binom{\tau}{i}. \quad (1)$$

Note that the proposed scheme is inspired by the ordered statistics decoding (OSD) of linear codes [11], whose search takes place over the set of codewords corresponding to a Hamming sphere for a selected basis.

Fig. 1 shows the block error rate (BLER) performance of impulsive perturbation decoding according to the dimension of the perturbation space τ for the (96, 48) LDPC code [10]. The normalized min-sum (NMS) decoding is used for component BP decoding and a CRC-8 is employed. The perturbation space selection criterion proposed in Section III-B is combined with the impulsive perturbation. Note that s is dependent on the values of T and τ owing to (1). For each curve of a fixed T , the rightmost is the data point for τ being the largest and s the smallest. As τ is set smaller, s increases step-wise and the performance changes. It is observed that the best performance is obtained where τ is set to the largest. When T is smaller or equal to the code length, as 14, 62, and 96, the best performance is achieved at $\tau = T$ where $s = 1$. In addition, for $T = 4656$, performance is the best at the maximum possible $\tau = 96$ where $s = 2$. When τ is small as 14-16, an increase of T does not give much performance gain. Whereas $\tau = \log_2 T$ for afterburner, the numerical results for the proposed scheme suggest $\tau = \min(T, n)$, which leads to a much wider but shallower search.

B. New Perturbation Space Selection Criterion

Before the perturbation space is determined, the symbols should be ordered to find the symbols of low reliability. Additionally the symbols more sensitive to perturbation are preferred. In [9], nSCV was proposed and shown to be more effective for AugBP decoding. Inspired by nSCV, we propose a new criterion effective for the impulsive perturbation. The criterion is the number of mismatches between the signs of the extrinsic and posterior LLRs, which we call the 'number of sign mismatch of variable nodes (nSMV)'. Let $l_e^i(j)$ be the extrinsic LLR and $l^i(j)$ be the posterior LLR of the j -th variable node at the i -th iteration of component BP decoding. Then, nSMV is defined by $m_j = |\{i | l_e^i(j)l^i(j) < 0, 1 \leq$

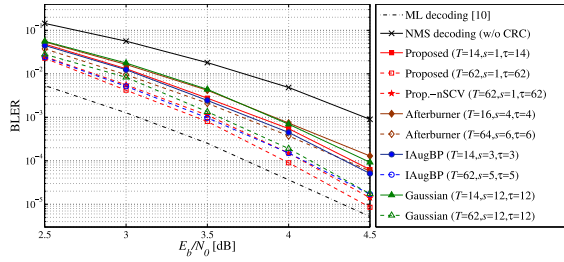


Fig. 2. Performance comparison for (96,48) code with CRC-8.

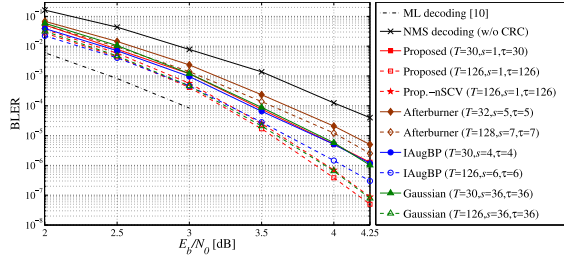


Fig. 3. Performance comparison for (192,96) code with CRC-8.

$i \leq I_{max}\}$, for the j -th variable node, $1 \leq j \leq n$. Symbols with more mismatches are considered less reliable. If the nSMVs are tied for different variable nodes, the intrinsic LLR magnitude is used for secondary rule. The proposed criterion achieves a performance gain of 0.12 dB for the (96,48) code, $T = 62$ and 0.07 dB for the (192,96) code, $T = 126$ at BLER = 10^{-4} versus nSCV criterion as shown in Figs. 2 and 3, respectively.

IV. NUMERICAL RESULTS

In this section, we evaluate the performance of the proposed scheme in comparison to the existing schemes: afterburner [7], IAUGBP [9], and the Gaussian perturbation [5]. All simulations assume the BPSK modulation over the AWGN channel, NMS decoder with a scaling factor of 0.78 and $I_{max} = 20$. Note that change of component BP decoder does not change the situation. The (96, 48) irregular codes [10] with a maximum variable node degree of four and (192, 96) irregular codes [10] with a maximum variable node degree of six are used. The codes are concatenated with the CRC-8 of $g(x) = x^8 + x^4 + x^3 + x^2 + 1$. The parameters of simulations are listed in Table I. As suggested in Section III, τ is chosen maximally as $\min(T, n)$ for the proposed scheme.

Fig. 2 exhibits the BLER performances of different perturbation decoding schemes for the (96,48) code. The proposed scheme achieves a performance gain of 0.7 dB for $T = 14$ and 1.1 dB for $T = 62$ at BLER = 10^{-4} over the plain NMS decoding and performs equally to or better than other schemes such as IAUGBP, afterburner, and Gaussian perturbation. A similar trend can be found in Fig. 3 for the (192,96) code. Overall, IAUGBP is as good as the proposed scheme for small T . Gaussian perturbation performs comparably for the (192,96) code without sign of error floor. However, the performance gain of the proposed scheme is notable for large T and IAUGBP and Gaussian perturbation are not parallelizable. Thus the proposed scheme is advantageous in both reliability and latency aspects. Fig. 4 compares the average number of

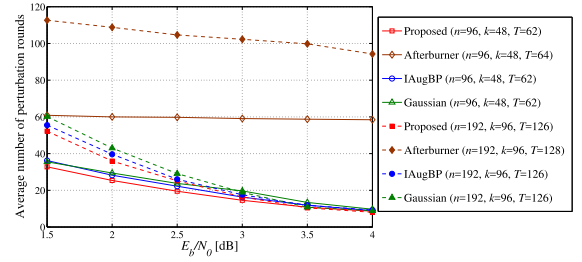


Fig. 4. Average number of perturbation rounds for the second phase.

perturbation rounds for rather large values of T . The proposed scheme has a 5-15% lower complexity than IAUGBP and Gaussian perturbation in the range of $E_b/N_0 = 1.5$ –3.0 dB.

V. CONCLUSION

In this letter, we investigated multi-round BP decoding with perturbation, that improves the performance of BP decoding for short LDPC codes. We first presented a common framework which can represent the existing algorithms as a special case, and provides a base for fair comparison between multiple perturbation schemes. Then, we proposed the impulsive perturbation scheme and the nSMV criterion for perturbation space selection. The numerical results also indicate that the perturbation space needs to be maximized and the number of perturbed symbols minimized for effective search for a successful perturbation pattern. The impulsive perturbation with an optimized τ outperforms the existing schemes. The second phase of the proposed scheme is parallelizable for high-speed decoding suitable to low latency applications.

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